1 Luhn's Algorithm implementation

The Luhn's algorithm also mod 10 algorithm, is a widely used checksum created by IBM scientist Hans Luhn in 1954 to validate credit card numbers, IMEI numbers and ensuring their accuracy.

Steps:

1. Start from right to left. Suppose we have a card number 123456. Starting from left to right, aka, reversing the order of card number, we will have 654321.

2. Double every even digit of the reversed number. If the result is greater than 9, then add the first and second digit using the formula $1 + x$ where x is the last digit. For example, if the sum is 12, the last digit is 2, so adding them up we will have $1 + 2 = 3$. Better still subtract9 from the resultant doubling i.e $(12 - 9 = 3)$.

3. Sum all the odd number together with the resultant sum of the even numbers.

4. Final step is to divide by 10 and check if the remainder is 0 then the card is valid else the card is invalid.

Now let us compute the above example:

$$CardNumber = 123456$$

$$ReversedNumber = 654321$$

$$DouledEven = 5^2 + 3^2 + 1^2 = 10$$

$$\rightarrow 10 + 6 + 1 = 1 + 6 + 1 = 8$$

$$DoubleOdd = 6 + 4 + 2 = 12$$

$$TotalSum = 20$$

$$TotalSum\%10 = 0$$

Thus a valid card number.